Arquitectura de Robots

Denavit-Hartenberg (DH) Method

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1 Denavit-Hartenberg (DH) Method

- Denavit-Hartenberg (DH) Method

Denavit-Hartenberg (DH) Method

- Proposed in 1955 by Denavit and Hartenberg.
- Systematic method to establish coordinate systems $\{S_i\}$ for each link.
- Enables kinematic equations for articulated chains.
- Uses 4 geometric transformations between links:
  1. Rotation θ_i about z_{i-1}
  2. Translation d_i along z_{i-1}
  3. Translation a_i along x_i
  4. Rotation α_i about x_i

DH Parameters

1. Rotation θ_i about z_{i-1}
2. Translation d_i along z_{i-1}
3. Translation a_i along x_i
4. Rotation α_i about x_i

Since the matrix product is not commutative, the transformations must be carried out in the indicated order. Transformation matrix between links:

$$^{i-1}A_i = R(z_{i-1}, \theta_i) \cdot \text{Trans}(z_{i-1}, d_i) \cdot \text{Trans}(x_i, a_i) \cdot R(x_i, \alpha_i)$$

DH Parameters frames

1 Denavit-Hartenberg (DH) Method

Transformations

$$^{i-1}A_i = R(z_{i-1}, \theta_i) \cdot \text{Trans}(z_{i-1}, d_i) \cdot \text{Trans}(x_i, a_i) \cdot R(x_i, \alpha_i)$$

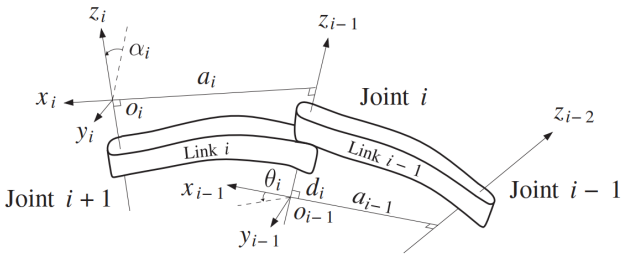

Figure 1: DH frame assignment.

DH Parameters frames Z_i

1 Denavit-Hartenberg (DH) Method

Transformations

$$^{i-1}A_i = R(z_{i-1}, \theta_i) \cdot \text{Trans}(z_{i-1}, d_i) \cdot \text{Trans}(x_i, a_i) \cdot R(x_i, \alpha_i)$$

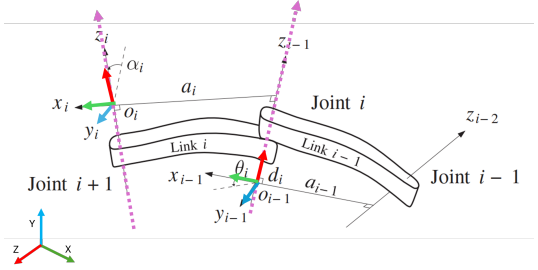

Figure 2: Z_i and Z_{i-1}

DH Parameters frames θ_i

1 Denavit-Hartenberg (DH) Method

Transformations

$$^{i-1}A_i = R(z_{i-1}, \theta_i) \cdot \text{Trans}(z_{i-1}, d_i) \cdot \text{Trans}(x_i, a_i) \cdot R(x_i, \alpha_i)$$

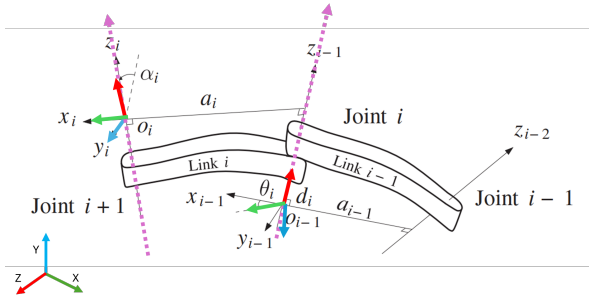

Figure 3: θ_i

DH Parameters frames d_i

1 Denavit-Hartenberg (DH) Method

Transformations

$$^{i-1}A_i = R(z_{i-1}, \theta_i) \cdot \text{Trans}(z_{i-1}, d_i) \cdot \text{Trans}(x_i, a_i) \cdot R(x_i, \alpha_i)$$

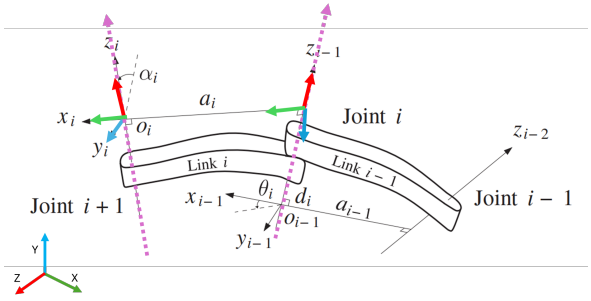

Figure 4: d_i

DH Parameters frames a_i

1 Denavit-Hartenberg (DH) Method

Transformations

$$^{i-1}A_i = R(z_{i-1}, \theta_i) \cdot \text{Trans}(z_{i-1}, d_i) \cdot \text{Trans}(x_i, a_i) \cdot R(x_i, \alpha_i)$$

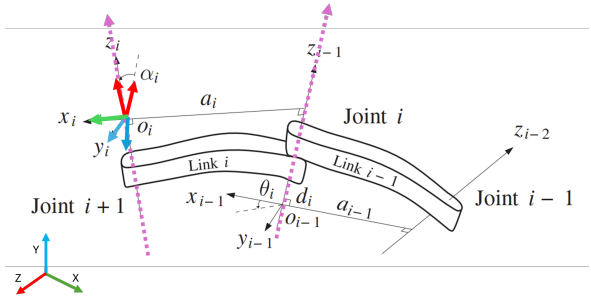

Figure 5: a_i

DH Parameters frames α_i

1 Denavit-Hartenberg (DH) Method

Transformations

$$^{i-1}A_i = R(z_{i-1}, \theta_i) \cdot \text{Trans}(z_{i-1}, d_i) \cdot \text{Trans}(x_i, a_i) \cdot R(x_i, \alpha_i)$$

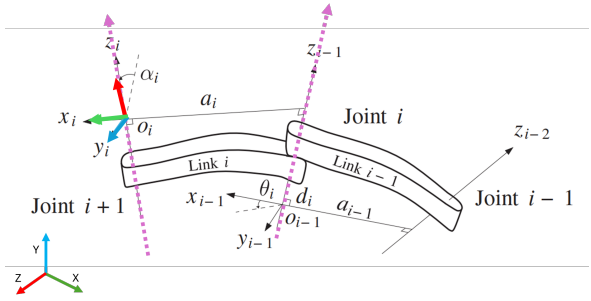

Figure 6: α_i

DH Transformation Matrix

1 Denavit-Hartenberg (DH) Method

Transformation matrix between links:

$$^{i-1}A_i = R(z_{i-1}, \theta_i) \cdot \text{Trans}(z_{i-1}, d_i) \cdot \text{Trans}(x_i, a_i) \cdot R(x_i, \alpha_i)$$

$$^{i}_{i-1}A = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & 0 \\ s\theta_i & c\theta_i & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & a_i \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & c\alpha_i & -s\alpha_i & 0 \\ 0 & s\alpha_i & c\alpha_i & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (1)$$

Resulting homogeneous matrix:

$$^{i-1}A_i = \begin{bmatrix} \cos\theta_i & -\cos\alpha_i \sin\theta_i & \sin\alpha_i \sin\theta_i & a_i \cos\theta_i \\ \sin\theta_i & \cos\alpha_i \cos\theta_i & -\sin\alpha_i \cos\theta_i & a_i \sin\theta_i \\ 0 & \sin\alpha_i & \cos\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Conditions

1 Denavit-Hartenberg (DH) Method

Conditions

DH1: The axis x_i must be perpendicular to Z_{i-1}
DH2 : The axis x_i intersects Z_{i-1}

$$ {}^{i-1}A_i = \begin{bmatrix} \cos\theta_i & -\cos\alpha_i \sin\theta_i & \sin\alpha_i \sin\theta_i & a_i \cos\theta_i \\ \sin\theta_i & \cos\alpha_i \cos\theta_i & -\sin\alpha_i \cos\theta_i & a_i \sin\theta_i \\ 0 & \sin\alpha_i & \cos\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

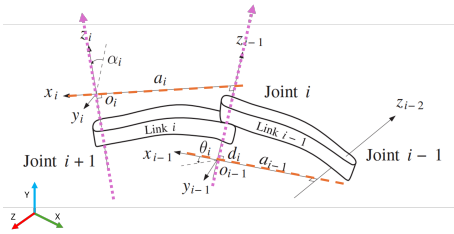

Figure 7: Conditions

DH Algorithm step 1

- 1. Number the links starting with 1 (first moving link in the chain) and ending with n (last moving link). The fixed base of the robot shall be numbered as link 0.

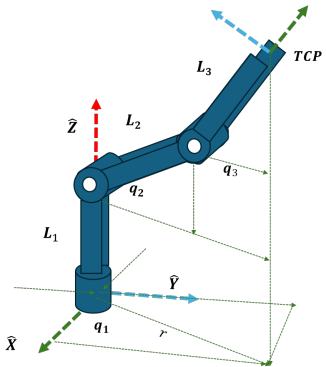

Figure 8: Step 1

DH Algorithm step 2

- 1. Number the links starting with 1 (first moving link in the chain) and ending with n (last moving link). The fixed base of the robot shall be numbered as link 0.
- 2. Number each joint starting with 1 (the one corresponding to the first degree of freedom) and ending in n.

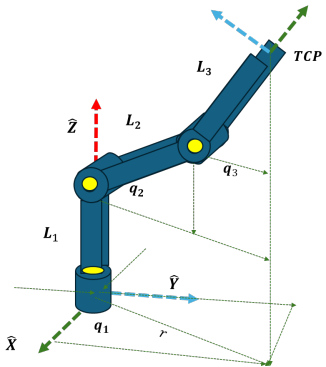

Figure 9: Step 2

DH Algorithm step 3

- 3. Locate the axis of each joint. If it is rotating, the axis will be its own axis of rotation; If it is prismatic, it will be the axis along which the displacement occurs.

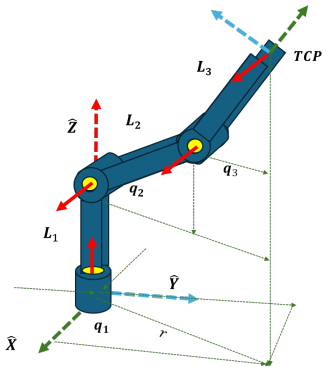

Figure 10: Step 3

DH Algorithm step 4

- 4. For i from 0 to $n-1$, place the axis z_i on the axis of the joint $i+1$.

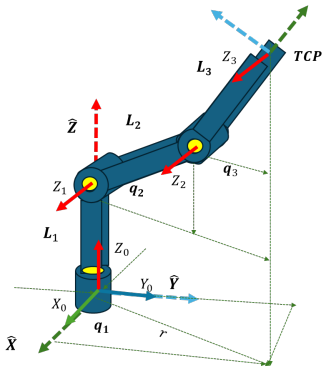

Figure 11: Step 4

DH Algorithm step 5

- 4. For i from 0 to $n-l$, place the axis z_1 on the axis of the joint $i+1$.
- 5. Place the system origin of the base S_o at any point on the axis Z_o. The axes X_o and Y_o shall be so positioned as to form a right-handed system with Z_o

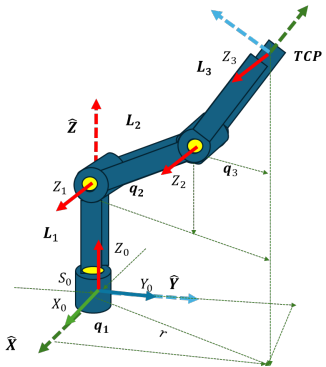

Figure 12: Step 5

DH Algorithm step 6

- 6. For i from 1 to $n-l$, locate the system S_i (solidarity to the link i) at the intersection of the axis Z_i with the normal line common to Z_{i-1} y Z_i

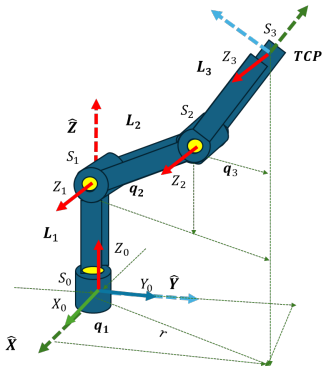

Figure 13: Step 6

DH Algorithm

6. For i from 1 to $n - l$, locate the system[Si] (solidarity to the link i) at the intersection of the axis Z_i with the normal line common to Z_{i-1} y Z_i

- If both axes intersect, S_i would be at the cut-off point.
- If they were parallel S_i it would be located in the joint $i + l$.

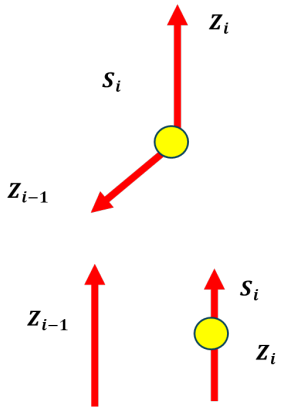

Figure 14: Z axes intersect

DH Algorithm step 7

- 7. Place x_i on the common normal line to Z_{i+1} and Z_i.

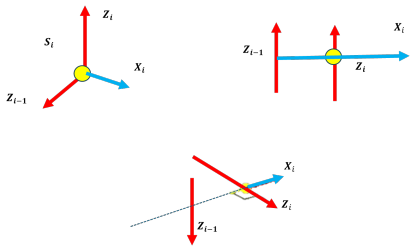

Figure 15: Step 7, Z axes

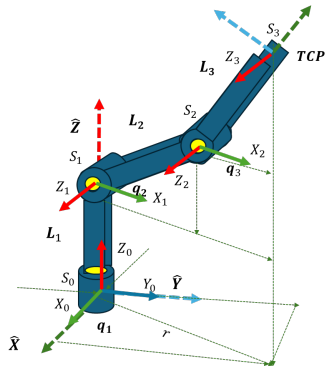

Figure 16: Step 7

DH Algorithm step 8

- 8. Place Y_i so that it forms a right-handed system with X_i and Z_i. (dextrorotatory)

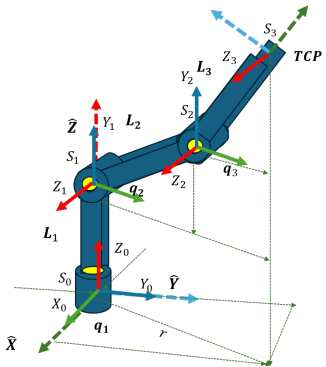

Figure 17: Step 8

DH Algorithm step 9

- 9. Place the system S_n at the end of the robot so that Z_n matches the direction of Z_{n-1} and X_n is normal to Z_{n-1} and Z_n

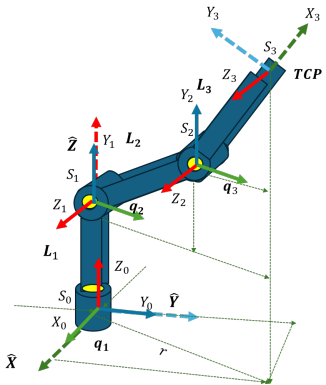

Figure 18: Step 9

DH Algorithm step 10, 11 , 12 , 13

- 10. Get θ_i as the angle that must be rotated around Z_{i-1} so that X_{i-1} and X_i are parallel.
- 11. Obtain d_i as the distance, measured along Z_{i-1}, that would have to be displaced S_{i-1} so that X_i and X_{i-1} were aligned.
- 12. Obtain a_i as the distance measured along X_i (which would now coincide with x...) that the new S_{i-1} would have to be displaced so that its origin coincided with S_i.
- 13. Obtain α_i as the angle that would have to be rotated around X_i (which would now coincide with X_{i-1}), so that the new S_{i-1} would coincide totally with S_i.

DH Algorithm step 14, 15, 16

- **14.** Obtain the defined transformation matrices A_i.
- **15.** Obtain the transformation matrix that relates the base frame to the end-effector frame T: ${}^0A_1, {}^1A_2, \ldots, {}^{n-1}A_n$.
- **16.** The matrix T defines the orientation (rotation submatrix) and position (translation submatrix) of the end-effector with respect to the base as a function of the n joint coordinates.

Coordinate systems to convert

- θ_i rotation about Z_{n-1} required to parallel X_{n-1} and X_n
- d_i distance from the center of plot $n-1$ to the center of frame n, along the axis Z_{n-1}
- a_i distance from the center of plot $n-1$ to the center of frame n, along the axis X_n
- α_i rotation about X_n required to parallel Z_{n-1} and Z_n

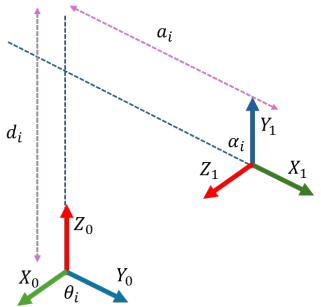

Figure 19: Coordinate systems to convert

Rotation θ_i about Z_i

- θ_i rotation about Z_{n-1} required to parallel X_{n-1} and X_n
- d_i distance from the center of plot $n-1$ to the center of frame n, along the axis Z_{n-1}
- a_i distance from the center of plot $n-1$ to the center of frame n, along the axis X_n
- α_i rotation about X_n required to parallel Z_{n-1} and Z_n

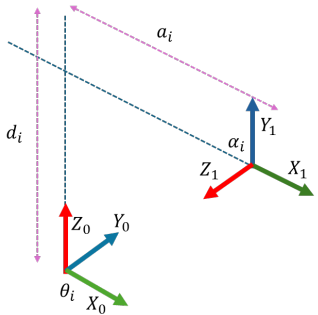

Figure 20: Rotation θ_i about Z_i

Translation d_i along Z_i

- θ_i rotation about Z_{n-1} required to parallel X_{n-1} and X_n
- d_i distance from the center of plot $n-1$ to the center of frame n, along the axis Z_{n-1}
- a_i distance from the center of plot $n-1$ to the center of frame n, along the axis X_n
- α_i rotation about X_n required to parallel Z_{n-1} and Z_n

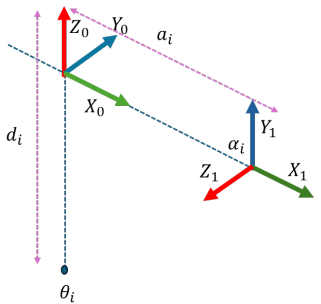

Figure 21: Translation d_i along Z_i

Translation a_i along X_i

- θ_i rotation about Z_{n-1} required to parallel X_{n-1} and X_n
- d_i distance from the center of plot $n-1$ to the center of frame n, along the axis Z_{n-1}
- a_i distance from the center of plot $n-1$ to the center of frame n, along the axis X_n
- α_i rotation about X_n required to parallel Z_{n-1} and Z_n

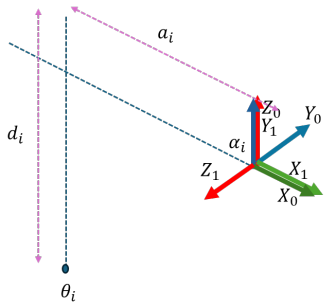

Figure 22: Traslation a_i along X_i

Rotation α_i about X_{i+1}

- θ_i rotation about Z_{n-1} required to parallel X_{n-1} and X_n
- d_i distance from the center of plot $n-1$ to the center of frame n, along the axis Z_{n-1}
- a_i distance from the center of plot $n-1$ to the center of frame n, along the axis X_n
- α_i rotation about X_n required to parallel Z_{n-1} and Z_n

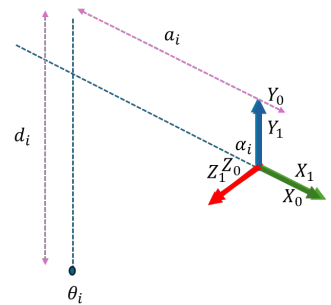

Figure 23: Rotation α_i about X_{i+1}

Section 1.1

Example

Robot RRR (DH Convention)

1 Denavit-Hartenberg (DH) Method

- θ_i: Rotation about z_{i-1}
- d_i: Translation along z_{i-1}
- a_i: Translation along x_i
- α_i: Rotation about x_i

S_0	θ_i	d_i	a_i	α_i
S_1	q_1			
S_2				
S_3				

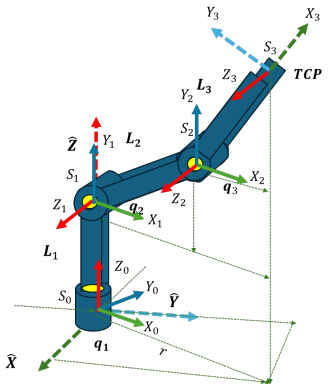

Figure 24: Robot 3DoF RRR

Robot RRR (DH Convention)

1 Denavit-Hartenberg (DH) Method

- θ_i: Rotation about z_{i-1}
- d_i: Translation along z_{i-1}
- a_i: Translation along x_i
- α_i: Rotation about x_i

S_0	θ_i	d_i	a_i	α_i
S_1	q_1	L_1		
S_2				
S_3				

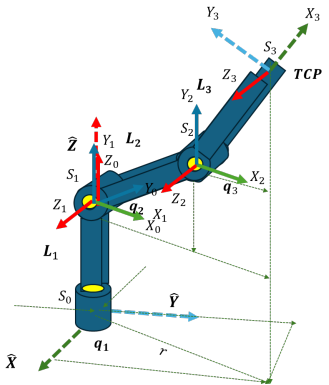

Figure 25: Robot 3DoF RRR

Robot RRR (DH Convention)

1 Denavit-Hartenberg (DH) Method

- θ_i: Rotation about z_{i-1}
- d_i: Translation along z_{i-1}
- a_i: Translation along x_i
- α_i: Rotation about x_i

S_0	θ_i	d_i	a_i	α_i
S_1	q_1	L_1	0	
S_2				
S_3				

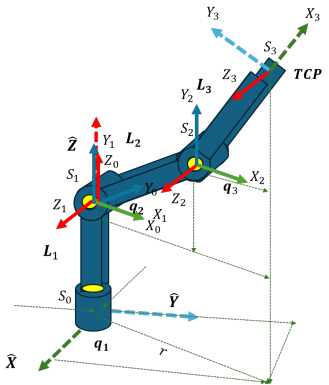

Figure 26: Robot 3DoF RRR

Robot RRR (DH Convention)

1 Denavit-Hartenberg (DH) Method

- θ_i: Rotation about z_{i-1}
- d_i: Translation along z_{i-1}
- a_i: Translation along x_i
- α_i: Rotation about x_i

S_0	θ_i	d_i	a_i	α_i
S_1	q_1	L_1	0	90°
S_2				
S_3				

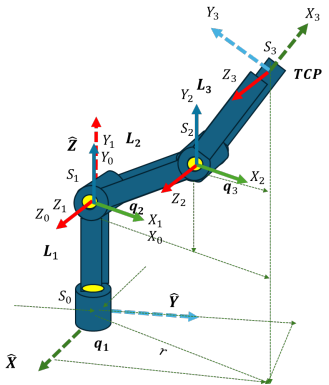

Figure 27: Robot 3DoF RRR

Robot RRR (DH Convention)

1 Denavit-Hartenberg (DH) Method

- θ_i: Rotation about z_{i-1}
- d_i: Translation along z_{i-1}
- a_i: Translation along x_i
- α_i: Rotation about x_i

S_0	θ_i	d_i	a_i	α_i
S_1	q_1	L_1	0	90°
S_2	q_2			
S_3				

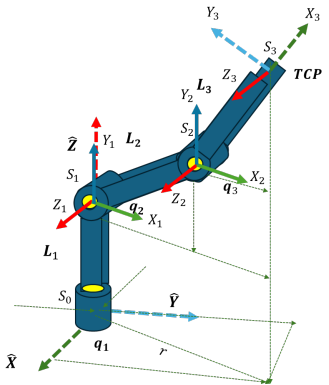

Figure 28: Robot 3DoF RRR

Robot RRR (DH Convention)

1 Denavit-Hartenberg (DH) Method

- θ_i: Rotation about z_{i-1}
- d_i: Translation along z_{i-1}
- a_i: Translation along x_i
- α_i: Rotation about x_i

S_0	θ_i	d_i	a_i	α_i
S_1	q_1	L_1	0	90°
S_2	q_2	0		
S_3				

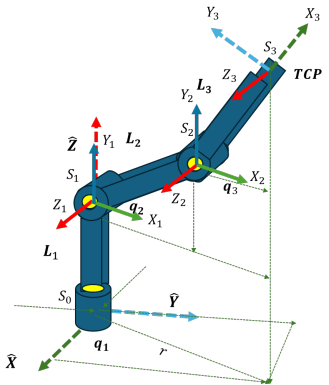

Figure 29: Robot 3DoF RRR

Robot RRR (DH Convention)

1 Denavit-Hartenberg (DH) Method

- θ_i: Rotation about z_{i-1}
- d_i: Translation along z_{i-1}
- a_i: Translation along x_i
- α_i: Rotation about x_i

S_0	θ_i	d_i	a_i	α_i
S_1	q_1	L_1	0	90°
S_2	q_2	0	L_2	
S_3				

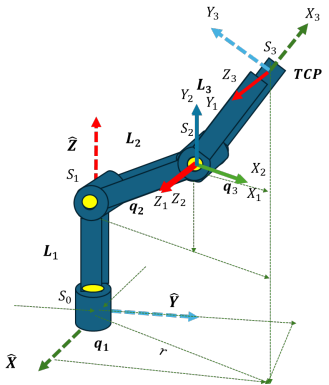

Figure 30: Robot 3DoF RRR

Robot RRR (DH Convention)

1 Denavit-Hartenberg (DH) Method

- θ_i: Rotation about z_{i-1}
- d_i: Translation along z_{i-1}
- a_i: Translation along x_i
- α_i: Rotation about x_i

S_0	θ_i	d_i	a_i	α_i
S_1	q_1	L_1	0	90°
S_2	q_2	0	L_2	0
S_3				

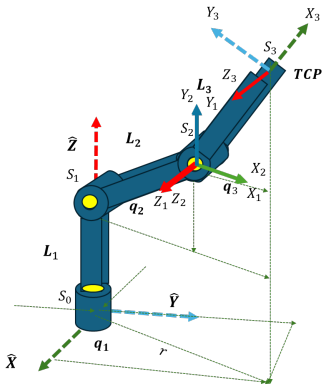

Figure 31: Robot 3DoF RRR

Robot RRR (DH Convention)

1 Denavit-Hartenberg (DH) Method

- θ_i: Rotation about z_{i-1}
- d_i: Translation along z_{i-1}
- a_i: Translation along x_i
- α_i: Rotation about x_i

S_0	θ_i	d_i	a_i	α_i
S_1	q_1	L_1	0	90°
S_2	q_2	0	L_2	0
S_3	q_3			

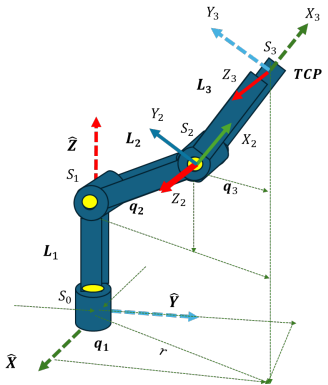

Figure 32: Robot 3DoF RRR

Robot RRR (DH Convention)

1 Denavit-Hartenberg (DH) Method

- θ_i: Rotation about z_{i-1}
- d_i: Translation along z_{i-1}
- a_i: Translation along x_i
- α_i: Rotation about x_i

S_0	θ_i	d_i	a_i	α_i
S_1	q_1	L_1	0	90°
S_2	q_2	0	L_2	0
S_3	q_3	0		

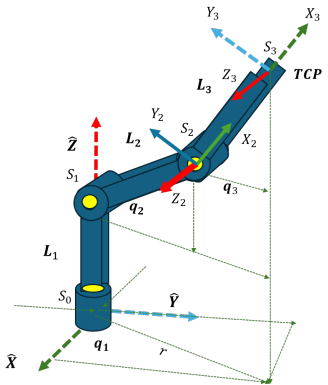

Figure 33: Robot 3DoF RRR

Robot RRR (DH Convention)

1 Denavit-Hartenberg (DH) Method

- θ_i: Rotation about z_{i-1}
- d_i: Translation along z_{i-1}
- a_i: Translation along x_i
- α_i: Rotation about x_i

S_0	θ_i	d_i	a_i	α_i
S_1	q_1	L_1	0	90°
S_2	q_2	0	L_2	0
S_3	q_3	0	L_3	

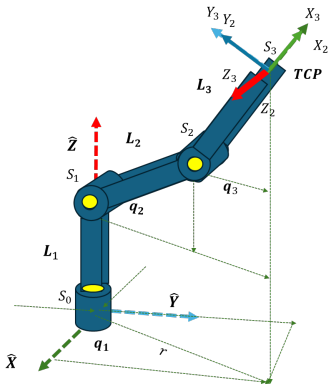

Figure 34: Robot 3DoF RRR

Robot RRR (DH Convention)

1 Denavit-Hartenberg (DH) Method

- θ_i: Rotation about z_{i-1}
- d_i: Translation along z_{i-1}
- a_i: Translation along x_i
- α_i: Rotation about x_i

S_0	θ_i	d_i	a_i	α_i
S_1	q_1	L_1	0	90°
S_2	q_2	0	L_2	0
S_3	q_3	0	L_3	0

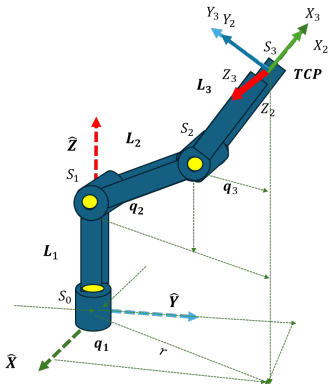

Figure 35: Robot 3DoF RRR

Robot RRR (DH Convention)

1 Denavit-Hartenberg (DH) Method

S_0	θ_i	d_i	a_i	α_i
S_1	q_1	L_1	0	90°
S_2	q_2	0	L_2	0
S_3	q_3	0	L_3	0

$$ {}^{i-1}A_i = \begin{bmatrix} \cos\theta_i & -\cos\alpha_i \sin\theta_i & \sin\alpha_i \sin\theta_i & a_i \cos\theta_i \\ \sin\theta_i & \cos\alpha_i \cos\theta_i & -\sin\alpha_i \cos\theta_i & a_i \sin\theta_i \\ 0 & \sin\alpha_i & \cos\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^{0}A_1 = \begin{bmatrix} cq_i & 0 & sq_1 & 0 \\ sq_i & 0 & -cq_1 & 0 \\ 0 & 1 & 0 & L_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad (2) $$

Robot RRR (DH Convention)

1 Denavit-Hartenberg (DH) Method

S_0	θ_i	d_i	a_i	α_i
S_1	q_1	L_1	0	90°
S_2	q_2	0	L_2	0
S_3	q_3	0	L_3	0

$$ {}^{i-1}A_i = \begin{bmatrix} \cos\theta_i & -\cos\alpha_i \sin\theta_i & \sin\alpha_i \sin\theta_i & a_i \cos\theta_i \\ \sin\theta_i & \cos\alpha_i \cos\theta_i & -\sin\alpha_i \cos\theta_i & a_i \sin\theta_i \\ 0 & \sin\alpha_i & \cos\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^{1}A_2 = \begin{bmatrix} cq_2 & -sq_2 & 0 & L_2 cq_2 \\ sq_2 & cq_2 & 0 & L_2 sq_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (3) $$

Robot RRR (DH Convention)

1 Denavit-Hartenberg (DH) Method

S_0	θ_i	d_i	a_i	α_i
S_1	q_1	L_1	0	90°
S_2	q_2	0	L_2	0
S_3	q_3	0	L_3	0

$$ {}^{i-1}A_i = \begin{bmatrix} \cos\theta_i & -\cos\alpha_i \sin\theta_i & \sin\alpha_i \sin\theta_i & a_i \cos\theta_i \\ \sin\theta_i & \cos\alpha_i \cos\theta_i & -\sin\alpha_i \cos\theta_i & a_i \sin\theta_i \\ 0 & \sin\alpha_i & \cos\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^{2}A_3 = \begin{bmatrix} cq_3 & -sq_3 & 0 & L_3 cq_3 \\ sq_3 & cq_3 & 0 & L_3 sq_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (4) $$

Robot RRR (DH Convention)

1 Denavit-Hartenberg (DH) Method

$$^{0}A_3 = ^{0}A_1 \cdot ^{1}A_2 \cdot ^{2}A_3$$

$$^{0}A_3 = \begin{bmatrix} cq_1 & 0 & sq_1 & 0 \\ sq_1 & 0 & -cq_1 & 0 \\ 0 & 1 & 0 & L_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} cq_2 & -sq_2 & 0 & L_2cq_2 \\ sq_2 & cq_2 & 0 & L_2sq_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} cq_3 & -sq_3 & 0 & L_3cq_3 \\ sq_3 & cq_3 & 0 & L_3sq_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (5)$$

Robot RRR (DH Convention)

1 Denavit-Hartenberg (DH) Method

$$
{}^{0}A_3 = \begin{bmatrix} cq_2cq_3cq_1 - cq_1sq_2sq_3 & -cq_2cq_1sq_3 - cq_3cq_1sq_2 & sq_1 & P_x \\ cq_2cq_3sq_1 - sq_2sq_3sq_1 & -cq_2sq_3sq_1 - cq_3sq_2sq_1 & -cq_1 & P_y \\ cq_2sq_3 + cq_3sq_2 & cq_2cq_3 - sq_2sq_3 & 0 & P_z \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$
Px = L_2cq_2cq_1 + L_3cq_2cq_3cq_1 - L_3cq_1sq_2sq_3
$$

$$
Py = L_2cq_2sq_1 + L_3cq_2cq_3sq_1 - L_3sq_2sq_3sq_1
$$

$$
Pz = L_2sq_2 + L_3cq_2sq_3 + L_3cq_3sq_2 + L_1
$$

Robot RRR (DH Convention)

1 Denavit-Hartenberg (DH) Method

$$\cos(a+b) = \cos a \cos b - \sin a \sin b \tag{6}$$

$$\sin(a+b) = \sin a \cos b + \cos a \sin b \tag{7}$$

$$c_2c_3 - s_2s_3 = c_{23}, \quad \text{where } c_{23} = \cos(q_2+q_3). \tag{8}$$

$$c_2s_3 + c_3s_2 = s_{23}, \quad \text{where } s_{23} = \sin(q_2+q_3). \tag{9}$$

$$^0A_3 = \begin{bmatrix} c_{23}c_1 & -s_{23}c_1 & s_1 & c_1(L_2c_2 + L_3c_{23}) \\ c_{23}s_1 & -s_{23}s_1 & -c_1 & s_1(L_2c_2 + L_3c_{23}) \\ s_{23} & c_{23} & 0 & L_2s_2 + L_3s_{23} + L_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

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Q&A

Thank you for your attention

Rodriguez, J. Fundamentos De Robotica Antonio Barrientos.
Craig, J. J. (2006). Introduction to robotics. Pearson Educacion..
Spong, M. W., Hutchinson, S., & Vidyasagar, M. (2020). Robot modeling and control. John Wiley & Sons.